

ANSFORD MANUFACTURING GROUP

Process Capability Study Report

Part / drawing:	PB-1140 pump body — drawing PB-1140 rev E
Characteristic:	Overall length 48.500 mm ± 0.250
Specification:	LSL 48.250 mm / USL 48.750 mm
Study period:	02/02/2026 – 13/03/2026
Sampling:	30 subgroups × 5 consecutive parts
Prepared by:	Priya Nair, Quality Engineer
Approved by:	Sarah Bright, Quality Manager

Results

Grand mean, $\bar{X}$	48.508 mm
Mean range, $\bar{R}$	0.098 mm
$\hat{\sigma} = \bar{R}/d_2$	0.0421 mm
Cp	1.98
Cpk	1.91
Meets ≥ 1.33	YES

Submitted to Harlow Fluid Systems on form QF114, 20/03/2026 — accepted 27/03/2026 (R. Hardy).